COMPATIBLE SYSTEMS OF PERFECT COMPLEXES AND ENDOMORPHISMS ASSEMBLE INTO A LIMIT COMPLEX WITH COMPATIBLE TRACES

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Allowed standard prerequisites (editorial).

- A perfect complex over the local ring Z/l^nZ is represented by a finite complex of finite free modules with differentials landing in l times the next term, unique up to isomorphism. invoked at proof lines 2613-2618. Provides the canonical representatives K_n^bullet and their uniform boundedness.
- The trace of an endomorphism of a bounded complex over Q_l computes as the alternating sum of traces on cohomology. invoked at proof lines 2680-2687. Identifies the limit trace with the trace on H^i(K_infinity tensor Q_l).

Source reference. Original theorem 03V2: sources/stacks-trace/pinned-original-source.tex, source lines 2523–2523. Cited at proof lines 2686-2688 to record that this computation also finishes the proof of theorem-C; navigational reference at the close of the span.

Lemma 0.2. *Suppose we have $K_n \in D_{perf}(\mathbf{Z}/\ell^n\mathbf{Z})$, $\pi_n : K_n \to K_n$ and isomorphisms $\varphi_n : K_{n+1} \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^{n+1}\mathbf{Z}} \mathbf{Z}/\ell^n\mathbf{Z} \to K_n$ compatible with π_{n+1} and π_n. Then*

1. *the elements $t_n = Tr(\pi_n|_{K_n}) \in \mathbf{Z}/\ell^n\mathbf{Z}$ form an element $t_\infty = \{t_n\}$ of $\mathbf{Z}_\ell$,*
2. *the $\mathbf{Z}_\ell$-module $H^i_\infty = \lim_n H^i(k_n)$ is finite and is nonzero for finitely many i only, and*
3. *the operators $H^i(\pi_n) : H^i(K_n) \to H^i(K_n)$ are compatible and define $\pi^i_\infty : H^i_\infty \to H^i_\infty$ satisfying*

$$\sum (-1)^i Tr(\pi^i_\infty|_{H^i_\infty \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell}) = t_\infty.$$

Proof. Since $\mathbf{Z}/\ell^n\mathbf{Z}$ is a local ring and K_n is perfect, each K_n can be represented by a finite complex $K_n^\bullet$ of finite free $\mathbf{Z}/\ell^n\mathbf{Z}$-modules such that the map $K_n^p \to K_n^{p+1}$ has image contained in ℓK_n^{p+1}. It is a fact that such a complex is unique up to isomorphism. Moreover π_n can be represented by a morphism of complexes $\pi_n^\bullet$:

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$K_n^\bullet \to K_n^\bullet$ (which is unique up to homotopy). By the same token the isomorphism $\varphi_n : K_{n+1} \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^{n+1}\mathbf{Z}} \mathbf{Z}/\ell^n\mathbf{Z} \to K_n$ is represented by a map of complexes

$$\varphi_n^\bullet : K_{n+1}^\bullet \otimes_{\mathbf{Z}/\ell^{n+1}\mathbf{Z}} \mathbf{Z}/\ell^n\mathbf{Z} \to K_n^\bullet.$$

In fact, $\varphi_n^\bullet$ is an isomorphism of complexes, thus we see that

- there exist $a, b \in \mathbf{Z}$ independent of n such that $K_n^i = 0$ for all $i \notin [a, b]$, and
- the rank of K_n^i is independent of n.

Therefore, the module $K_\infty^i = \lim_n \{K_n^i, \varphi_n^i\}$ is a finite free $\mathbf{Z}_\ell$-module and $K_\infty^\bullet$ is a finite complex of finite free $\mathbf{Z}_\ell$-modules. By induction on the number of nonzero terms, one can prove that $H^i(K_\infty^\bullet) = \lim_n H^i(K_n^\bullet)$ (this is not true for unbounded complexes). We conclude that $H_\infty^i = H^i(K_\infty^\bullet)$ is a finite $\mathbf{Z}_\ell$-module. This proves *ii*. To prove the remainder of the lemma, we need to overcome the possible non-commutativity of the diagrams

$$\begin{array}{ccc} K_{n+1}^\bullet & \xrightarrow{\varphi_n^\bullet} & K_n^\bullet \\ \downarrow{\scriptstyle \pi_{n+1}^\bullet} & & \downarrow{\scriptstyle \pi_n^\bullet} \\ K_{n+1}^\bullet & \xrightarrow[\varphi_n^\bullet]{} & K_n^\bullet. \end{array}$$

However, this diagram does commute in the derived category, hence it commutes up to homotopy. We inductively replace $\pi_n^\bullet$ for $n \geq 2$ by homotopic maps of complexes making these diagrams commute. Namely, if $h^i : K_{n+1}^i \to K_n^{i-1}$ is a homotopy, i.e.,

$$\pi_n^\bullet \circ \varphi_n^\bullet - \varphi_n^\bullet \circ \pi_{n+1}^\bullet = dh + hd,$$

then we choose $\tilde{h}^i : K_{n+1}^i \to K_{n+1}^{i-1}$ lifting h^i. This is possible because K_{n+1}^i free and $K_{n+1}^{i-1} \to K_n^{i-1}$ is surjective. Then replace $\pi_n^\bullet$ by $\tilde{\pi}_n^\bullet$ defined by

$$\tilde{\pi}_{n+1}^\bullet = \pi_{n+1}^\bullet + d\tilde{h} + \tilde{h}d.$$

With this choice of $\{\pi_n^\bullet\}$, the above diagrams commute, and the maps fit together to define an endomorphism $\pi_\infty^\bullet = \lim_n \pi_n^\bullet$ of $K_\infty^\bullet$. Then part *i* is clear: the elements $t_n = \sum (-1)^i \mathrm{Tr}\left(\pi_n^i|_{K_n^i}\right)$ fit into an element t_∞ of $\mathbf{Z}_\ell$. Moreover

$$\begin{aligned} t_\infty &= \sum (-1)^i \mathrm{Tr}_{\mathbf{Z}_\ell}(\pi_\infty^i|_{K_\infty^i}) \\ &= \sum (-1)^i \mathrm{Tr}_{\mathbf{Q}_\ell}(\pi_\infty^i|_{K_\infty^i \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell}) \\ &= \sum (-1)^i \mathrm{Tr}(\pi_\infty|_{H^i(K_\infty^\bullet \otimes \mathbf{Q}_\ell)}) \end{aligned}$$

where the last equality follows from the fact that $\mathbf{Q}_\ell$ is a field, so the complex $K_\infty^\bullet \otimes \mathbf{Q}_\ell$ is quasi-isomorphic to its cohomology $H^i(K_\infty^\bullet \otimes \mathbf{Q}_\ell)$. The latter is also equal to $H^i(K_\infty^\bullet) \otimes_{\mathbf{Z}} \mathbf{Q}_\ell = H_\infty^i \otimes \mathbf{Q}_\ell$, which finishes the proof of the lemma, and also that of Theorem 03V2. $\square$